Xinrui(Richard) Yi

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Education

Northeastern University — **San Jose, CA**
M.S. in Computer Science; ***GPA: 3.9*** — *Sep 2023 – May 2026 (Expected)*

- **Relevant Courses:**Machine Learning, Natural Language Processing, Distributed Systems, Cloud Computing
- **Leadership:** Career Peer Advisor, Teaching Assistant

University at Buffalo — **Buffalo, NY**
B.S. & M.S. in Civil Engineering; ***CS-Related GPA: 4.0/4.0*** — *Feb 2017 – May 2021*

Skills

Languages: Python, Java, C++, JavaScript, TypeScript, Go
AI/ML: PyTorch, TensorFlow, LangChain, vLLM, Claude Code, Cursor
Frameworks & Tools: Spring Boot, Django, Node.js, Flask, Flutter, Docker, Kubernetes, Git,
Databases & Cloud: MySQL, MongoDB, Supabase, PostgreSQL, Redis, Kafka, AWS (S3, EC2, Lambda), GCP

Experience

Astreas Performance *(RAG, FastAPI, MLOps, Docker, Terraform)* — **Mountain View, CA**
ML Engineer Intern — *Jun 2025 – Aug 2025*

- Architected and scaled a **production-ready RAG** system for a health optimization platform, processing **8,000+** daily queries from **5,000+** beta users with **99.5%** uptime on Google Cloud Run.
- Ran systematic evaluations on RAG pipeline modules through **AutoRAG**, identifying latency bottlenecks (2.3s avg response time) and low retrieval accuracy as key areas for improvement.
- Implemented **Weaviate** as the vector database with **HNSW indexing** and an **inverted index** for **hybrid search** (dense + sparse retrieval), combined with an **overlapping chunking** strategy and **cross-encoder reranker**, reducing query latency from 2.3s to 1.4s and boosting retrieval relevance from 0.73 to 0.89 on **NDCG@10**.
- Integrated **PathRAG** (graph-based RAG) to structure and query **multi-domain** health data across a **50GB+** dataset, improving recommendation accuracy from 72% to 90% as measured by **Recall** and **F1 score**.
- Deployed the full stack with **Terraform**-managed **Docker** containers on **GCP**, leveraging **auto-scaling** across multiple instances with **FastAPI** and **Streamlit**, supporting 200 concurrent users while reducing infrastructure costs by **30%**.

Voc.ai *(Prometheus, Locust, Redis, Kafka, AWS(EC2, ElastiCache))* — **San Jose, CA**
AI Agent Engineer Intern — *Jan 2025 – Apr 2025*

- Built and deployed **AI-powered customer service agents** for enterprise clients, developing **database query tools** for order lookup and integrating **knowledge bases** for enterprise-specific information.
- Configured **Prometheus** dashboards to monitor backend API latency and throughput, and ran **Locust** load tests to identify performance bottlenecks under concurrent user traffic.
- Implemented a **Redis** caching layer for frequently accessed customer and order data, reducing average query response time from 280ms to 95ms.
- Integrated **Kafka** message queues to process customer inquiry events asynchronously, enabling non-blocking request handling across multiple AI agents with a **99.97%** message delivery rate.
- Deployed application infrastructure on **AWS**, scaling from 850 to 1,200 concurrent users while maintaining **99.8%** uptime.

China Railway Major Bridge Reconnaissance & Design Institute *(Java, MySQL)*
Design & Software Engineer — *Jul 2021 – Jul 2023*

- Developed an internal **workflow automation** tool using **Java** for engineers to generate technical drafts from predefined parameters and templates, improving design cycle efficiency by **30%**.
- Built a keyword-based drawing retrieval module integrated with AutoCAD workflows, reducing manual search time for engineering documents across large-scale bridge and infrastructure projects.
- Collaborated with engineers in a **cross-function team** to determine technical specifications.

Project

BiteWise: AI-Powered Meal Planning Android App | ***Vertex AI, LangChain, Firebase*** — **(Github Link)**

- **Top 30 Finalist** TED AI hackathon (largest UN Sustainability Goal Hackathon)
- Designed and developed a meal tracking and planning application using **Vertex AI** framework and **Firebase** backend, enabling users to track dietary habits and receive AI-powered recipe recommendations.
- Integrated **Gemini API** to provide personalized recipe suggestions and **OpenAI DALL·E 3 API** to generate recipe photo.
- Utilized **Chain of Thought** through **LangChain** to minimize **98%** AI hallucination.